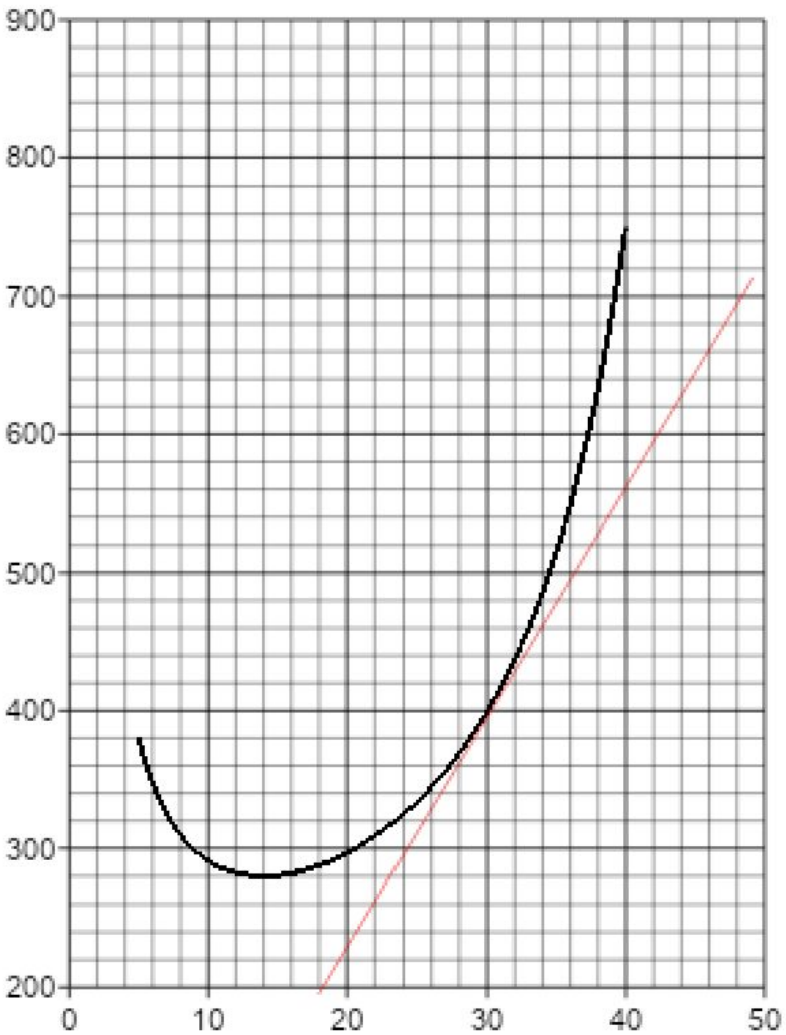
5 **(a)** The potential of both charge near their surface is positive B1

(b) From x = 5 cm to about 14 cm, (resultant) electric field (strength) is towards B and decreasing in magnitude. B1

(Resultant) electric field (strength) is zero at *x* ~14 cm B1

From x H 14 cm to x = 40 cm, (resultant) electric field (strength) is towards A and increasing in magnitude. B1

(c) show construction and correct calculation using $\Delta V / \Delta x$ M1
(0.21,250), (0.44,630)
E = 1650 N C^{-1}



(d) Consider x = 0.30 m, E = 1650 NC ^{1} (or any other appropriate point)

$$-\frac{1}{4\pi\varepsilon_0}\left(\frac{Q_A}{0.30^2}\right)+\frac{1}{4\pi\varepsilon_0}\left(\frac{8.0\times10^{-9}}{(0.50-0.30)^2}\right)=1650$$ M1

Q_A = 1.48 nC or 1.5 nC A1

(e) Minimum kinetic energy required to reach A
= gain in electric potential energy from surface of A to neutral point
= e ($V_{\text{neutral point}} - V_{\text{surface of B}}$)

Change in potential = (280 – 750) C1

From energy conservation / loss in k.e. = gain in e.p.e. M1

$\frac{1}{2}(9.11 \times 10^{-31})v^2 = e(750 - 280) = 7.52 \times 10^{-17}$

v = 1.28 × 10^7 m s^{-1} or 1.3 × 10^7 m s^{-1} A1